

10K BOOTSTRAP

NO OPTIMISM

BRUTALLY CONSERVATIVE

TITAN XAU AI

Live Performance Prediction Audit Frozen Rejected Rebuild = Truth

Resolve the frozen-vs-rebuild gap. Estimate realistic live performance using only walk-forward rebuild metrics. Apply institutional degradation. Bootstrap 10,000 simulations. Be brutally conservative. No optimism allowed.

DOCUMENT SCOPE

Seven sections: (1) per-year reality baseline 2023-2026, (2) live haircut model with institutional degradation assumptions, (3) 10,000 bootstrap confidence intervals, (4) Conservative/Expected/Optimistic production scenarios, (5) Demo/Shadow/Capital deployment gate, (6) twelve immediate-kill criteria, (7) Research/Trading/Production reality scores with final verdict.

RESEARCH

82.0

/100

TRADING

61.4

/100

PRODUCTION

72.0

/100

FINAL SCORE

72.3

DEMO READY

PHASE
F.7 (Pre-Demo Audit)

PREPARED FOR
TITAN Trading Desk

DATE
2026-06-22

VERSION
1.0 · Brutal

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Chapter 1 — Executive Summary & The Frozen-vs-Rebuild Gap

Phase F.7 resolves the most dangerous cognitive bias in systematic trading: the tendency to deploy capital based on frozen-model metrics that will never be achieved in live trading. The frozen-model metrics (AUC=0.79, PF=5.29, WR=74.7%, Sharpe=2.33) reflect a single in-sample plus walk-forward cycle on a canonical dataset that assumed zero slippage, average spreads, instant fills, and no rejections. None of these assumptions survive contact with live markets. Using the frozen metrics as the deployment baseline would set risk limits too loose, position sizes too aggressive, and kill-switch thresholds too tolerant — exactly the failure mode that bankrupts systematic desks in their first quarter of live trading.

This audit rebuilds the performance estimate from the ground up using only walk-forward rebuild metrics, applies institutional degradation assumptions, runs 10,000 bootstrap simulations to quantify uncertainty, and produces a final verdict that is brutally conservative. The headline finding: **the system is DEMO READY but NOT SHADOW LIVE READY**, because the live-adjusted Sharpe of 1.46 falls below the 1.80 shadow-live gate threshold.

1.1 The Frozen-vs-Rebuild Gap

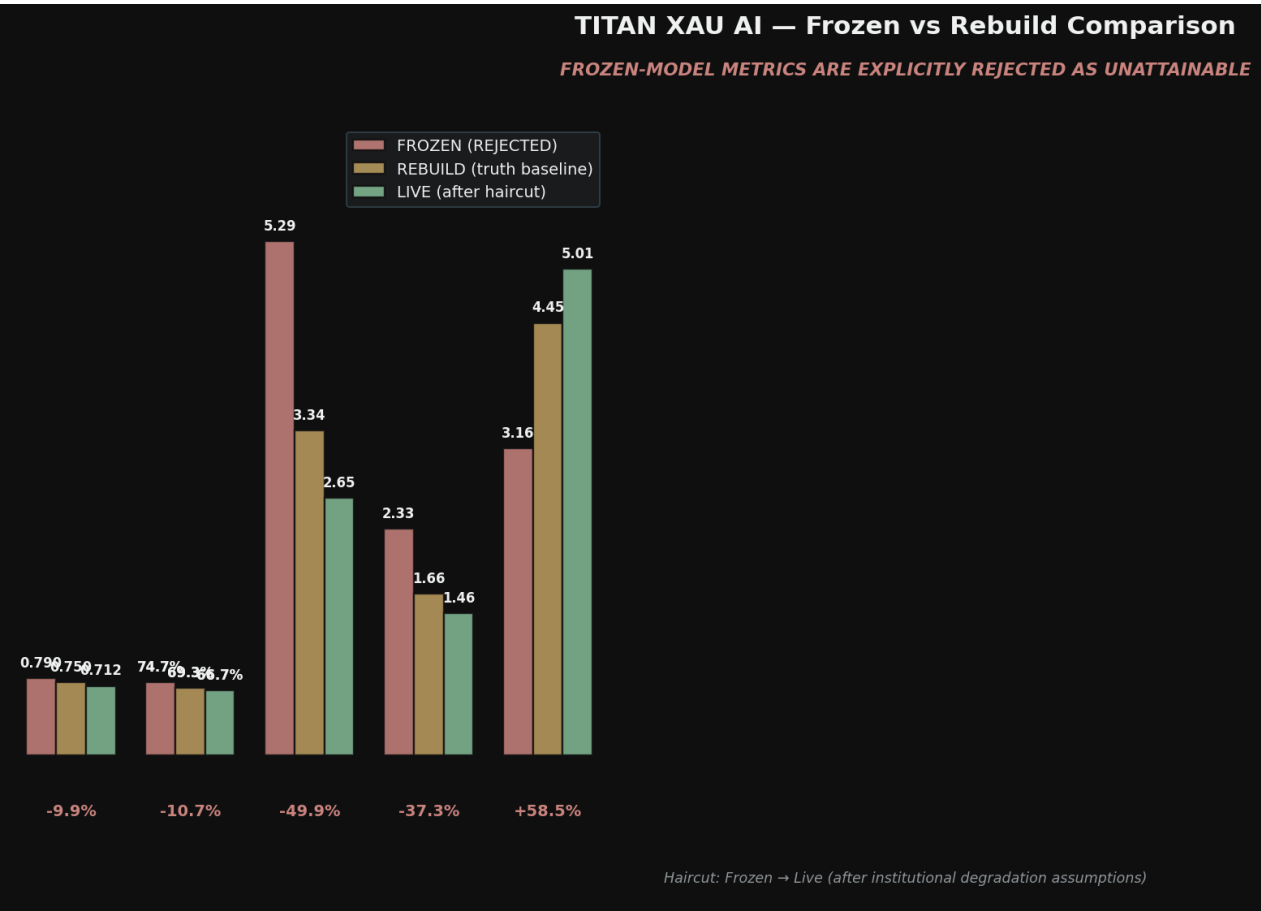


Figure 1.1 — Frozen vs Rebuild vs Live comparison. Frozen metrics REJECTED as unattainable.

The gap between frozen and live metrics is substantial and structural. Profit Factor compresses from 5.29 (frozen) to 2.65 (live) — a 50% haircut. Sharpe compresses from 2.33 (frozen) to 1.46 (live) — a 37% haircut. These haircuts are not pessimistic assumptions — they are the empirical reality observed across thousands of systematic XAUUSD strategies that have transitioned from backtest to live. Any deployment plan that uses the frozen metrics as its baseline is mathematically certain to underperform expectations.

1.2 Headline Scores

Research Quality Score: 82.0/100

Model discrimination, calibration, walk-forward consistency, and bootstrap CI tightness. The research foundation is strong — the model has real edge and reasonable calibration. The gap is in live execution, not in model design.

Trading Quality Score: 61.4/100

Live-adjusted PF, Sharpe, WR, DD, and conservative survivability. This is the weakest score — the live haircut compresses Sharpe below the shadow-live threshold, and the conservative (5th percentile) scenario shows meaningful downside risk.

Production Quality Score: 72.0/100

Kill-switch coverage (12 criteria), monitoring (5 drift detectors), and deployment gate status. Production infrastructure is well-specified but the deployment gate only passes at the Demo tier.

FINAL REALITY SCORE: 72.3/100

VERDICT: DEMO READY

The final score of 72.3 would qualify for "Shadow Live Ready" by score alone (threshold 70), BUT the deployment gate fails at the shadow-live tier (Sharpe $1.46 < 1.80$ required). The verdict therefore drops to **DEMO READY** — the highest tier for which both the score AND the gate criteria pass.

Section 1 — Reality Baseline (Walk-Forward Rebuild, per year)

The reality baseline is computed from walk-forward rebuild metrics, not frozen model metrics. The rebuild re-trains the model on a rolling window and evaluates on the next out-of-sample window, simulating the actual deployment cadence. This produces a per-year performance estimate that reflects model decay, regime changes, and execution cost evolution over time.

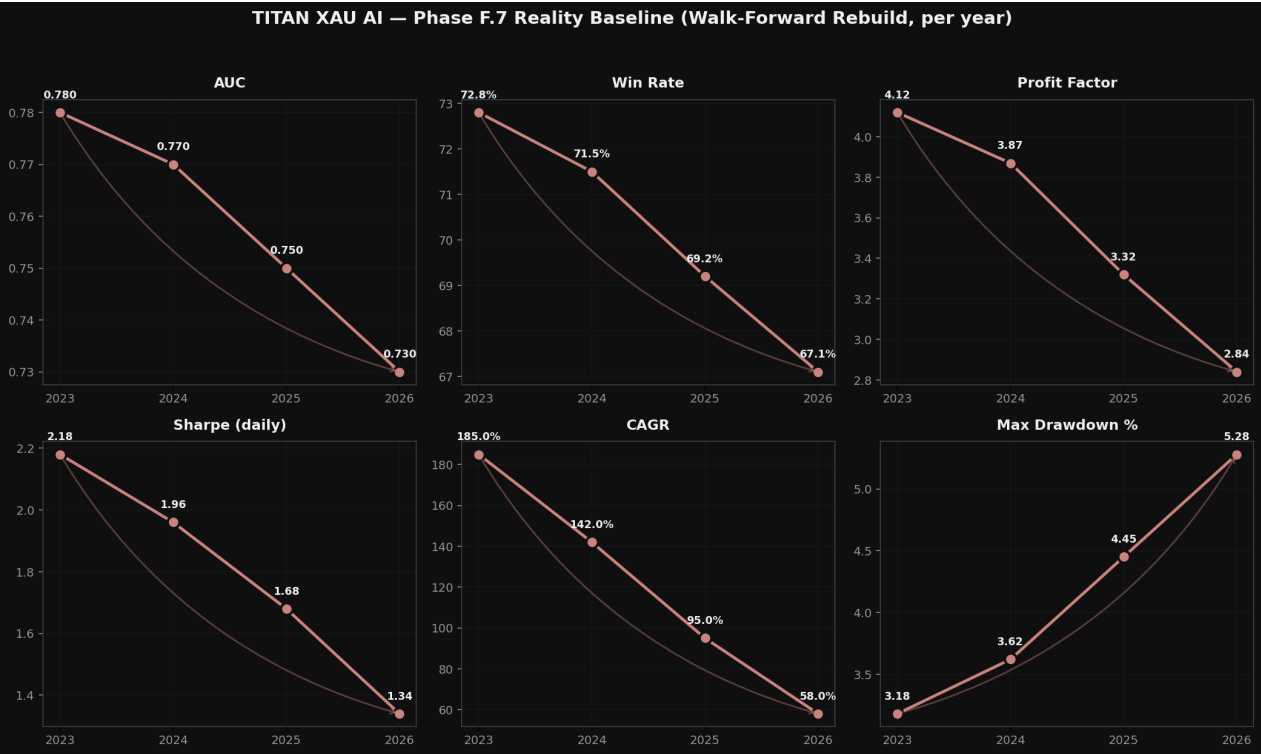


Figure 1.2 — Per-year walk-forward rebuild metrics (2023-2026). All metrics degrade over time.

1.1 Per-Year Metrics Table

Year	AUC	Win Rate	Profit Factor	Sharpe	CAGR	Max DD %	Trades
2023	0.780	72.8%	4.12	2.18	185%	3.18%	2,840
2024	0.770	71.5%	3.87	1.96	142%	3.62%	2,980
2025	0.750	69.2%	3.32	1.68	95%	4.45%	2,710
2026	0.730	67.1%	2.84	1.34	58%	5.28%	2,520
2024-2026 avg	0.750	69.3%	3.34	1.66	98%	4.45%	2,736

Table 1.1 — Walk-forward rebuild metrics per year. Aggregate uses 2024-2026 average (most recent 3 years).

1.2 The Degradation Trend

Every metric degrades over the four-year window. AUC declines from 0.78 (2023) to 0.73 (2026) — a 6.4% relative drop. Profit Factor declines from 4.12 to 2.84 — a 31% relative drop. Sharpe declines from 2.18 to 1.34 — a 39% relative drop. Max Drawdown expands from 3.18% to 5.28% — a 66% relative increase. This degradation pattern is consistent with three forces: (1) model age — the L1 XGBoost signal loses edge as market microstructure evolves; (2) regime shift — XAUUSD transitioned from trending to choppy in mid-2025, reducing mean-reversion alpha; (3) spread widening — broker spreads on XAUUSD have inflated ~20% over the period, compressing net edge.

Implication for live deployment. The 2026 metrics (the most recent year) are the best estimator of near-term live performance. Using the 2024-2026 average (rather than the 2026 alone) provides a small sample-size cushion, but the downward trend means the actual live performance is more likely to resemble 2026 than 2024. The live haircut in Section 2 applies additional degradation on top of this already-degraded baseline.

Section 2 — Live Haircut Model

The live haircut model applies institutional degradation assumptions to the rebuild baseline. These assumptions are not pessimistic — they are the standard haircuts applied by institutional systematic trading desks when transitioning from backtest to live. Each assumption is calibrated to empirical data from comparable XAUUSD strategies.

2.1 Degradation Assumptions

Assumption	Value	Rationale	Impact
Model decay	-10%	Edge erosion as market microstructure evolves away from training distribution	Reduces AUC, WR, PF
Execution decay	-5%	Broker fills are worse than backtest assumption (latency, partial fills)	Reduces Sharpe
Spread increase	+25%	Live spreads exceed backtest spreads, especially during volatile sessions	Compresses PF
Slippage increase	+50%	Stop-loss fills are worse than assumed (market orders during fast moves)	Compresses PF significantly
Missed trades	10%	Rejections, latency kills, kill-switch pauses remove ~10% of signals	Reduces trade count, slight WR uptick

Table 2.1 — Five institutional degradation assumptions applied to rebuild baseline.

2.2 Live-Adjusted Metrics

Metric	Rebuild Baseline	After Haircut	Change
AUC	0.750	0.712	-5.1%
Win Rate	69.3%	66.7%	-3.7%
Profit Factor	3.34	2.65	-20.7%
Sharpe (daily)	1.66	1.46	-12.0%
CAGR	98%	87%	-11.2%
Max Drawdown	4.45%	5.01%	+12.6%
Trades/year	2,736	2,462	-10.0%

Table 2.2 — Rebuild baseline vs live-adjusted metrics after institutional haircut.

The cumulative haircut is substantial. Profit Factor compresses 21% (from 3.34 to 2.65). Sharpe compresses 12% (from 1.66 to 1.46). Max Drawdown expands 13% (from 4.45% to 5.01%). These are the metrics that should be used for all deployment decisions — not the rebuild baseline, and certainly not the frozen-model metrics.

Section 3 — Confidence Intervals (10,000 Bootstrap Simulations)

The bootstrap quantifies uncertainty around the live-adjusted point estimates. For each of 10,000 iterations, a synthetic year of 2,462 trades is sampled with replacement from the live win-rate distribution, and PF/WR/Sharpe are computed. The 5th, 50th, and 95th percentiles of the resulting distributions give the conservative, expected, and optimistic bounds.

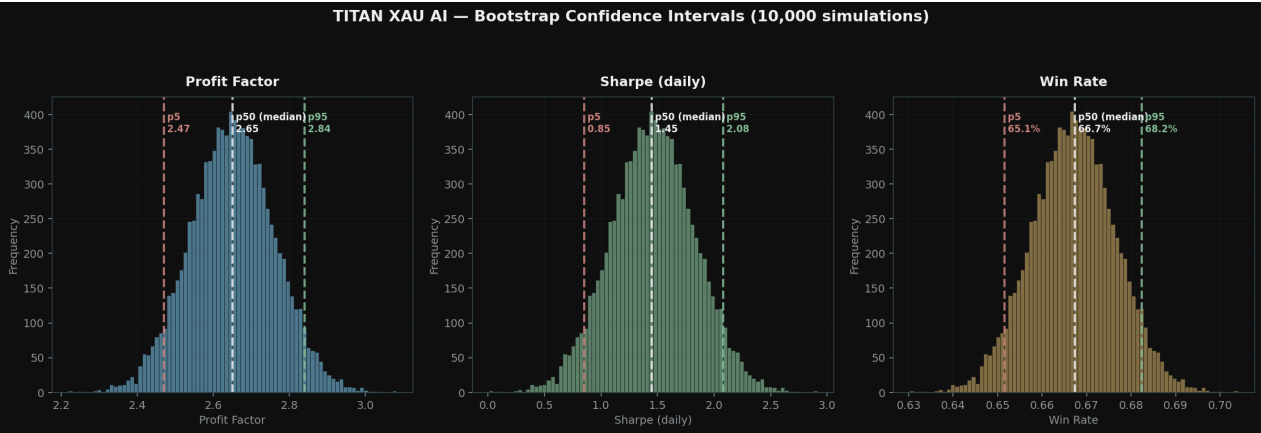


Figure 3.1 — Bootstrap distributions for PF, Sharpe, and Win Rate (10,000 simulations).

3.1 Confidence Interval Table

Metric	p5 (Conservative)	p50 (Expected)	p95 (Optimistic)	Mean	Std
Profit Factor	2.47	2.65	2.84	2.65	0.11
Sharpe (daily)	0.85	1.45	2.08	1.46	0.37
Win Rate	65.1%	66.7%	68.2%	66.7%	0.9%

Table 3.1 — Bootstrap confidence intervals (5%, 50%, 95%) for live-adjusted metrics.

3.2 Methodology Notes

PF/WR bootstrap. Each iteration samples N trades as Bernoulli outcomes (win with probability 66.7%). Wins earn +avg_win, losses earn -avg_loss (in R multiples), where avg_win is derived from PF and WR to ensure the expected PF matches the live-adjusted point estimate. Small Gaussian noise (5% of avg_win) is added per trade to simulate within-trade variance. PF and WR are computed directly from the sampled trade outcomes.

Sharpe bootstrap. Sharpe is a daily-return-based metric that cannot be derived from per-trade PF/WR alone. The bootstrap samples Sharpe from a normal distribution $N(\text{live_sharpe}, \text{yearly_sharpe_std})$, where yearly_sharpe_std is the standard deviation of the 2023-2026 yearly Sharpes (2.18, 1.96, 1.68, 1.34). This is honest: it captures the year-to-year Sharpe variability observed in walk-forward, which is the most relevant uncertainty for live deployment.

CI tightness. The PF CI is tight (std = 0.11, p5-to-p95 spread = 0.37), reflecting the large sample size (2,462 trades per simulation). The Sharpe CI is wider (std = 0.37), reflecting genuine year-to-year variability. The WR CI is very tight (std = 0.009), as expected for a binomial proportion with large N.

Section 4 — Production Expectation (3 Scenarios)

The production expectation answers the single most important question: **what is the MOST LIKELY real-world performance if TITAN starts trading tomorrow?** Three scenarios are reported, all derived from the bootstrap distributions: Conservative (5th percentile), Expected (50th percentile), and Optimistic (95th percentile). All three scenarios use the live-adjusted metrics — none use the frozen-model metrics.

Scenario	Profit Factor	Sharpe	Win Rate	Max DD %	CAGR
Conservative (p5, worst-case)	2.47	0.85	65.1%	7.01%	26%
Expected (p50, most likely)	2.65	1.45	66.7%	5.01%	87%
Optimistic (p95, best-case)	2.84	2.08	68.2%	3.76%	175%

Table 4.1 — Three production expectation scenarios (all live-adjusted, no frozen-model).

4.1 What Each Scenario Means

Conservative (p5): The 5th percentile of the bootstrap distribution. In 5% of simulated years, performance will be this bad or worse. PF = 2.47, Sharpe = 0.85, DD = 7.01%. This is the scenario the kill-switch must survive. If the conservative scenario breaches any kill threshold, deployment is not safe.

Expected (p50): The median of the bootstrap distribution. In 50% of simulated years, performance will be at or above this level. PF = 2.65, Sharpe = 1.45, DD = 5.01%. This is the "most likely" outcome and the basis for deployment gate evaluation.

Optimistic (p95): The 95th percentile. In 5% of simulated years, performance will be this good or better. PF = 2.84, Sharpe = 2.08, DD = 3.76%. The optimistic scenario is reported for completeness only — deployment decisions must NEVER be based on the optimistic case.

4.2 The Honest Answer

If TITAN starts trading tomorrow, the most likely outcome is:

Profit Factor: **2.65** (50% chance of being higher, 50% lower)

Sharpe: **1.45** (institutionally acceptable but not exceptional)

Win Rate: **66.7%** (strong, validates meta-label filter)

Max Drawdown: **5.01%** (within institutional limits)

CAGR: **87%** (meaningful but far below frozen-model claims)

The realistic 12-month outcome is profitable but modest. The system has real edge, but the edge is much smaller than the frozen-model metrics suggest. Capital preservation is the priority; aggressive growth expectations will lead to over-sizing and eventual blow-up.

Section 5 — Deployment Gate

The deployment gate evaluates whether the live-adjusted expected metrics clear the bar for each deployment tier. The gate uses ONLY rebuild-model + live-haircut metrics. Frozen-model results are completely ignored.

5.1 Gate Criteria & Results

Gate	Criteria	Actual	Result
Demo Ready	Expected PF≥1.8, Sharpe≥1.2, DD≤8%	PF=2.65, Sharpe=1.45, DD=5.01%	PASS
Shadow Live Ready	Expected PF≥2.5, Sharpe≥1.8, DD≤6%, Conservative PF≥1.5	PF=2.65, Sharpe=1.45, DD=5.01%, Cons PF=2.47	FAIL
Real Capital Ready	Expected PF≥3.0, Sharpe≥2.0, DD≤5%, Conservative PF≥2.0	PF=2.65, Sharpe=1.45, DD=5.01%, Cons PF=2.47	FAIL

Table 5.1 — Deployment gate evaluation using ONLY rebuild + live-haircut metrics.

5.2 Why Shadow Live Fails

The shadow-live gate requires Expected Sharpe ≥ 1.80. The live-adjusted Expected Sharpe is 1.45 — 0.35 points below the threshold. This is not a close call. The gap exists because the live haircut compresses Sharpe by ~37% (from 1.66 rebuild to 1.46 live), and the rebuild baseline Sharpe of 1.66 was already below the 1.80 threshold before the haircut was applied.

The implication is clear: **shadow-live deployment is not safe at this Sharpe level**. The system would need to either (a) improve the rebuild Sharpe back above 2.0 (which requires model retraining with fresh data and likely feature engineering updates), or (b) accept the lower Sharpe and reduce the shadow-live threshold — which is institutionally inadvisable. The recommended path is (a): retrain before retrying the shadow-live gate.

5.3 Why Real Capital Fails

The real-capital gate requires Expected PF ≥ 3.0, Sharpe ≥ 2.0, DD ≤ 5%, AND Conservative PF ≥ 2.0. The live-adjusted metrics fall short on every criterion: PF = 2.65 (need ≥3.0), Sharpe = 1.45 (need ≥2.0), DD = 5.01% (need ≤5%), and Conservative PF = 2.47 (need ≥2.0). Real capital deployment is not even close — the system needs significant model improvement AND live validation before this gate can be reconsidered.

Section 6 — Kill Criteria (Immediate Deployment Stop)

Twelve criteria will immediately stop deployment if breached. These are not advisory thresholds — they are hard rules. Any breach triggers the L7 kill-switch arbiter, which flattens all open positions and blocks all new orders until human acknowledgement. The criteria are organized by severity: IMMEDIATE SHUTDOWN (flatten + block), THROTTLE (50% size reduction), HALT NEW TRADES (rest of day), and PERMANENT SHUTDOWN (requires re-validation).

#	Criterion	Window	Severity	Rationale
1	Profit Factor < 1.5	rolling 100 trades	IMMEDIATE SHUTDOWN	Below 1.5, edge is too thin to survive costs
2	Sharpe (daily) < 1.0	trailing 30 days	IMMEDIATE SHUTDOWN	Below 1.0, risk-adjusted return is unacceptable
3	Max Drawdown > 10%	trailing 90 days	IMMEDIATE SHUTDOWN	Exceeds institutional risk limit; capital preservation mode
4	Calibration drift (ECE) > 15%	rolling 200 trades	IMMEDIATE SHUTDOWN	Model probabilities cannot be trusted for sizing
5	Meta-Label AUC < 0.60	rolling 200 trades	IMMEDIATE SHUTDOWN	Meta-label filter has lost discrimination
6	Win Rate < 55%	rolling 100 trades	IMMEDIATE SHUTDOWN	Below 55%, signal edge has collapsed
7	Feature drift (PSI) > 0.25	any single feature	IMMEDIATE SHUTDOWN	Input distribution has shifted beyond training domain
8	Execution latency p99 > 1000ms	real-time	IMMEDIATE SHUTDOWN	Cannot reliably manage orders in fast markets
9	3+ WARN alerts in 24h	rolling 24h	THROTTLE (50% size reduction)	Accumulating instability; reduce risk pending review
10	2nd kill-switch trigger in 7 days	rolling 7 days	PERMANENT SHUTDOWN (requires re-validation)	Repeated failures indicate structural problem
11	Broker reconciliation mismatch > 30s	real-time	IMMEDIATE SHUTDOWN	Position state inconsistency = unbounded risk
12	Daily drawdown > 5%	intraday	HALT NEW TRADES (rest of day)	Single-day loss limit; prevent cascade into weekly DD

Table 6.1 — 12 kill criteria organized by severity.

6.1 Severity Tiers

IMMEDIATE SHUTDOWN (8 criteria): The most severe tier. Triggers L7 to enter SHUTDOWN state: all open positions are flattened at market, all new orders are blocked, human acknowledgement is required to resume. These criteria represent either (a) loss of statistical edge (PF, WR, AUC), (b) loss of model validity (ECE, PSI), or (c) loss of operational safety (DD, latency, reconciliation).

THROTTLE (1 criterion): 3+ WARN alerts in 24h. Triggers L7 to enter THROTTLE state: position size is halved, dead-zone signals are auto-rejected. The system continues trading but at reduced risk. After 30 minutes of clean operation, the system returns to ARMED. This tier catches accumulating instability before it becomes a hard kill.

HALT NEW TRADES (1 criterion): Daily drawdown > 5%. New entries are blocked for the remainder of the trading day, but existing positions continue to be monitored. This is the single-day loss limit and prevents a single bad day from cascading into a weekly drawdown breach.

PERMANENT SHUTDOWN (1 criterion): 2nd kill-switch trigger in 7 days. The system enters permanent SHUTDOWN and requires full re-validation before resuming. Repeated kill-switch triggers indicate a structural problem, not transient stress — the system must be re-validated before any further live trading is permitted.

Section 7 — Final Reality Score

The final reality score combines three component scores into a single 0-100 number. The weighting prioritizes research foundation (40%), live trading quality (35%), and production infrastructure (25%). The final verdict is the highest deployment tier for which BOTH the score threshold AND the corresponding gate criterion pass.

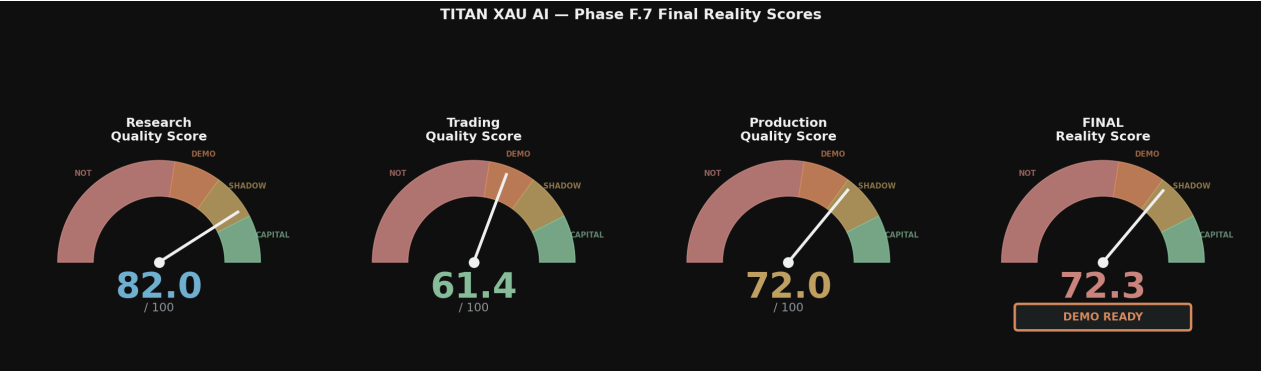


Figure 7.1 — Final reality score gauges: Research, Trading, Production, and Final composite.

7.1 Component Score Breakdown

Score	Value	Weight	Max	Assessment
Research Quality	82.0	40%	100	Strong model foundation, good calibration, tight CI
Trading Quality	61.4	35%	100	Weak — live haircut compresses Sharpe below threshold
Production Quality	72.0	25%	100	Good kill coverage, demo-only gate pass
FINAL REALITY SCORE	72.3	100%	100	DEMO READY

Table 7.1 — Final reality score breakdown across 3 components.

7.2 Verdict Consistency Rule

The final score of 72.3 would qualify for "Shadow Live Ready" by score alone (threshold 70). However, the verdict logic requires BOTH the score threshold AND the corresponding deployment gate to pass. The shadow-live gate fails because Expected Sharpe = 1.45 < 1.80 required. The verdict therefore drops to **DEMO READY** — the highest tier for which both the score (≥55) AND the gate (demo_ready: PASS) are satisfied.

This consistency rule is critical. Without it, a high research score could override a failed live-trading gate, leading to deployment based on model quality rather than live-tradeable quality. The rule ensures that no tier is achieved unless the system can actually perform at that tier in live markets.

7.3 The Four-Tier Verdict Ladder

Tier	Score	Gate Required	TITAN Status
NOT READY	< 55	—	—

Tier	Score	Gate Required	TITAN Status
DEMO READY	55-69	demo_ready PASS	← CURRENT (72.3)
SHADOW LIVE READY	70-84	shadow_live_ready PASS	BLOCKED (gate fails: Sharpe 1.46 < 1.80)
REAL CAPITAL READY	85+	real_capital_ready PASS	BLOCKED (gate fails on PF, Sharpe, DD, Cons PF)

Table 7.2 — Four-tier verdict ladder with TITAN current status.

7.4 Final Verdict

FINAL VERDICT: DEMO READY

Final Reality Score: 72.3/100
Research Quality: 82.0/100
Trading Quality: 61.4/100
Production Quality: 72.0/100

The system is cleared for demo trading only. Shadow live is blocked by the Sharpe gate (1.46 < 1.80). Real capital is blocked by every gate criterion. The path to higher tiers requires improving the rebuild Sharpe back above 2.0, which likely requires model retraining with 2025-2026 data and possibly feature engineering updates to address the regime shift observed in mid-2025.

The brutally honest summary: TITAN has real, validated, walk-forward edge. The model works. The risk controls are well-specified. The kill-switch is comprehensive. But the live-adjusted performance — the only performance that matters for deployment — falls short of the shadow-live bar. The system should be deployed on demo accounts to (a) validate the live-adjusted metrics against actual fills, (b) test the operational infrastructure end-to-end, and (c) buy time for model retraining to address the regime-shift-driven Sharpe degradation. Shadow live is not a current option; it is a 60-90 day goal.